

i) Necht' počet zab je z_n , počet konári k_n v roce n , z_0, k_0 jsou počáteční podmínky.

Ze zadání určíme vzorec:

$$\begin{aligned} z_{n+1} &= \frac{7}{10} z_n + \frac{k_n}{10000} \\ k_{n+1} &= \frac{13}{10} k_n - 500 z_n \end{aligned}$$

$$\text{Necht' } \bar{x}_n = \begin{pmatrix} z_n \\ k_n \end{pmatrix} \rightarrow \bar{x}_{n+1} = \begin{pmatrix} \frac{7}{10} & \frac{1}{10000} \\ -500 & \frac{13}{10} \end{pmatrix} \bar{x}_n = \bar{x}_{n+1} = \begin{pmatrix} \frac{7}{10} & \frac{1}{10000} \\ -500 & \frac{13}{10} \end{pmatrix}^n \bar{x}_0$$

ii) Diagonalizujeme matici $A = \begin{pmatrix} \frac{7}{10} & \frac{1}{10000} \\ -500 & \frac{13}{10} \end{pmatrix}$, aby se dobře počítalo.

Vlastní čísla A jsou $\left\{ \frac{6}{5}, \frac{4}{5} \right\}$,

$$M_{\frac{6}{5}} = \text{span} \left\{ \begin{pmatrix} 1 \\ 5 \cdot 10^3 \end{pmatrix} \right\}$$

$$M_{\frac{4}{5}} = \text{span} \left\{ \begin{pmatrix} 1 \\ 10^3 \end{pmatrix} \right\}$$

$$A^n = \begin{pmatrix} 1 & 1 \\ 5 \cdot 10^3 & 10^3 \end{pmatrix} \begin{pmatrix} \frac{6}{5} & 0 \\ 0 & \frac{4}{5} \end{pmatrix}^n \begin{pmatrix} 1 & 1 \\ 5 \cdot 10^3 & 10^3 \end{pmatrix}^{-1} = \frac{1}{4} \begin{pmatrix} 3 \cdot \left(\frac{4}{5}\right)^n - \left(\frac{6}{5}\right)^n & 10^3 \left(\left(\frac{6}{5}\right)^n - \left(\frac{4}{5}\right)^n\right) \\ 10^3 \left(3 \cdot \left(\frac{4}{5}\right)^n - 5 \cdot \left(\frac{6}{5}\right)^n\right) & 5 \left(\frac{4}{5}\right)^n - \left(\frac{6}{5}\right)^n \end{pmatrix}$$

iii) Dosadíme do vzorce pro $\bar{x}_n$.

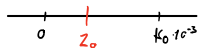
$$\begin{aligned} \bar{x}_n = A^n \bar{x}_0 &= \frac{1}{4} \begin{pmatrix} 3 \cdot \left(\frac{4}{5}\right)^n - \left(\frac{6}{5}\right)^n & 10^3 \left(\left(\frac{6}{5}\right)^n - \left(\frac{4}{5}\right)^n\right) \\ 10^3 \left(3 \cdot \left(\frac{4}{5}\right)^n - 5 \cdot \left(\frac{6}{5}\right)^n\right) & 5 \left(\frac{4}{5}\right)^n - \left(\frac{6}{5}\right)^n \end{pmatrix} \begin{pmatrix} z_0 \\ k_0 \end{pmatrix} = \frac{1}{4 \cdot 5^n} \begin{pmatrix} 20^3 \cdot 4^n - 20^3 \cdot 6^n + k_0 \cdot 10^3 \cdot 6^n - k_0 \cdot 10^3 \cdot 4^n \\ 20^3 \cdot 3 \cdot 4^n - 20^3 \cdot 5 \cdot 6^n + k_0 \cdot 5 \cdot 6^n - k_0 \cdot 4^n \end{pmatrix} \\ &= \frac{1}{4 \cdot 5^n} \begin{pmatrix} 6^n (10^3 k_0 - 20^3) + 4^n (3 \cdot 20^3 - 10^3 k_0) \\ 6^n (5 k_0 - 5 \cdot 10^3 z_0) + 4^n (3 \cdot 10^3 z_0 - k_0) \end{pmatrix} = \frac{1}{4} \begin{pmatrix} \left(\frac{6}{5}\right)^n (10^3 k_0 - 20^3) + \left(\frac{4}{5}\right)^n (3 \cdot 20^3 - 10^3 k_0) \\ \left(\frac{6}{5}\right)^n (5 k_0 - 5 \cdot 10^3 z_0) + \left(\frac{4}{5}\right)^n (3 \cdot 10^3 z_0 - k_0) \end{pmatrix} \end{aligned}$$

$$\frac{10^3}{4} k_0 - \frac{1}{4} 10^3$$

a) z_n i k_n jde do nekonečna.

$\checkmark 0 < z_0 < 10^3 k_0$ } podle prvního řádku

$5 k_0 > 5 \cdot 10^3 z_0 \Leftrightarrow z_0 < 10^3 k_0$ } podle druhého řádku
 $4 k_0 > 2 \cdot 10^3 z_0 \Leftrightarrow z_0 < 2 \cdot 10^3 k_0$



Zvolíme libovolné $k \in \mathbb{R}, k > 0$.

$$z_n = \left(\frac{6}{5}\right)^n (10^3 k_0 - 20^3) + \left(\frac{4}{5}\right)^n (3 \cdot 20^3 - 10^3 k_0) > \left(\frac{4}{5}\right)^n (10^3 k_0 - 20^3) > k$$

$$k_n = \left(\frac{6}{5}\right)^n (5 k_0 - 5 \cdot 10^3 z_0) + \left(\frac{4}{5}\right)^n (3 \cdot 10^3 z_0 - k_0) > \left(\frac{4}{5}\right)^n (5 k_0 - 5 \cdot 10^3 z_0) > k$$

Tedy podle desimice limity z_n i k_n diverguje $k \rightarrow \infty$.

Pro a) tedy platí $\left\{ \begin{pmatrix} z_0 \\ k_0 \end{pmatrix} \right\} = \left\{ \begin{pmatrix} t \\ 10^3 + t \end{pmatrix}, t \in \mathbb{R}, t \geq 0 \right\}$

b) $k_n < 0, n \in \mathbb{N}$.

Nejprve zvažme všechny situace, kdy $k_0 < 0$. To je zřejmé řešení.

Dále $k_0 = 0$. Pak ze vzorce: $k_1 < 0$. $\begin{pmatrix} x \\ t \end{pmatrix}, x \in \mathbb{R}, t \in \mathbb{R}, t \leq 0$. obdobně $\begin{pmatrix} t \\ x \end{pmatrix}, x \in \mathbb{R}, t \in \mathbb{R}, t \leq 0$.

Pak nechť $z_0 > 10^{-3} k_0 > 0$

$$\left(\frac{6}{5}\right)(5k_0 - 5 \cdot 10^3 z_0) < 0$$

$$\left(\frac{6}{5}\right)(3 \cdot 10^3 z_0 - k_0) < 0$$

$$\left(\frac{6}{5}\right)(5k_0 - 5 \cdot 10^3 z_0) \cdot \left(\frac{6}{5}\right)(3 \cdot 10^3 z_0 - k_0) < 0$$

$$\left(\frac{6}{5}\right)(3 \cdot 10^3 z_0 - k_0) < \left(\frac{6}{5}\right)(5 \cdot 10^3 z_0 - 5k_0)$$

$$\left(\frac{6}{5}\right)(3 \cdot 10^3 z_0 - k_0) < 0 < \left(\frac{6}{5}\right)(5 \cdot 10^3 z_0 - 5k_0)$$

což platí.

Naborec nechť $z_0 = 10^{-3} k_0 > 0$.

$$k_n = \left(\frac{6}{5}\right)^n (5k_0 - 5 \cdot 10^3 z_0) \cdot \left(\frac{6}{5}\right)^n (3 \cdot 10^3 z_0 - k_0) = \left(\frac{6}{5}\right)^n (3k_0 - k_0)$$

$$z \left(\frac{6}{5}\right)^n k_0 > 0, \text{ a } z \cdot \left(\frac{6}{5}\right)^n k_0 = 0, \text{ tedy}$$

Tedy tento případ nepatří do b).

Tedy $k_n < 0, n \in \mathbb{N}$, pro následující množinu $\bar{x}_0$:

$$\left\{ \begin{pmatrix} z_0 \\ k_0 \end{pmatrix} \right\} = \left\{ \begin{pmatrix} x \\ y \end{pmatrix}, x, y \in \mathbb{R}, x \leq 0 \vee y \leq 0 \right\} \cup \left\{ \begin{pmatrix} t \\ 10^{-3} - \varepsilon \end{pmatrix}, t, \varepsilon \in \mathbb{R}, t, \varepsilon > 0 \right\}$$